CHAPTER 6

Controller Design

In this chapter we introduce a number of techniques that can be used to design controllers. Our main goal is to understand how the location of the poles and zeros of the open-loop system influence the location of the poles of the closed-loop system. We start with a thorough study of second-order systems and culminate with a graphic tool known as *root locus*, in which the poles of the closed-loop system can be plotted in relation to the open-loop poles and zeros and the loop gain. Along the way we introduce derivative control and the ubiquitous proportional-integral-derivative controller.

6.1 Second-Order Systems

In our investigation of feedback systems we have had several encounters with second-order systems. Stable first- and second-order systems epitomize the basic behaviors of stable dynamic linear systems: exponentially decaying potentially oscillatory responses. Consider a second-order system with a characteristic polynomial in *canonical form*:

$$s^2 + 2\,\zeta\,\omega_n s + \omega_n^2 = 0, \qquad \omega_n > 0. \tag{6.1}$$

Most[1] second-order polynomials can be put in this form by an adequate choice of ω_n and ζ. We provide concrete examples later. The nature of the response of a second-order system is controlled by the location of the roots of the characteristic polynomial (6.1):

$$\begin{aligned} s^2 + 2\,\zeta\,\omega_n s + \omega_n^2 &= (s - s_1)(s - s_2), \\ s_1 = -\omega_n(\zeta + \sqrt{\zeta^2 - 1}), &\qquad s_2 = -\omega_n(\zeta - \sqrt{\zeta^2 - 1}). \end{aligned} \tag{6.2}$$

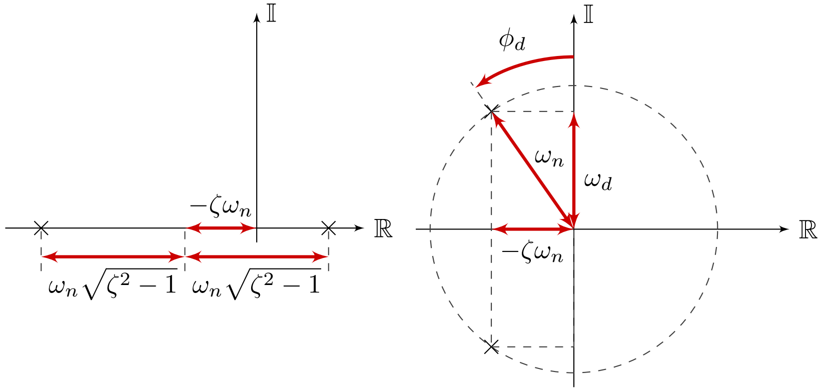
Note that the parameter ω_n only *scale* the roots, and that the parameter ζ controls whether the roots are real or complex-conjugate: if $|\zeta| \geq 1$ the roots are real; if $|\zeta| < 1$ they are complex conjugates.

When $|\zeta| \geq 1$ a second-order system has real roots and its response is the superposition of the response of two first-order systems. Indeed, the inverse

[1]The only second-order polynomials that are not encoded in (6.1) have the form:

$$s^2 + 2\,\zeta\,\omega_n s - \omega_n^2 = 0, \qquad \omega_n > 0.$$

Verify in **P6.1** that in this case the roots are always real, with one of the roots always positive.



(a) Real roots, $\zeta > 1$

(b) Complex roots, $0 < \zeta < 1$

Fig. 6.1: Location of the roots of the second-order equation (6.1); roots are marked with '$\times$'

Laplace transform of a second-order transfer-function with real poles has terms:

$$\mathcal{L}^{-1}\left\{\frac{k_1}{s-s_1}+\frac{k_2}{s-s_2}\right\} = k_1\,e^{s_1 t}+k_2\,e^{s_2 t}, \quad t \geq 0,$$

where the exact values of k_1 and k_2 depend on the zeros of the transfer-function and can be computed based on residues as in § 3. In the complex-plane, the roots are located on the real axis symmetrically about the point $s = -\zeta\omega_n$, as shown in Fig. 6.1a. If $\zeta > 1$ the roots are on the left-hand side of the complex plane, hence the associated transfer-function is asymptotically stable. If $\zeta < -1$ the roots are on the right-hand side of the complex plane, and the associated transfer-function is unstable.

When $|\zeta| < 1$ a second-order system has complex-conjugate roots and its response is oscillatory The inverse Laplace transform of such a second-order transfer-function contains the terms:

$$\mathcal{L}\left\{\frac{k}{s-(\zeta\omega_n - j\omega_d)}+\frac{k^*}{s-(\zeta\omega_n + j\omega_d)}\right\} = 2|k|e^{-\zeta\omega_n t}\cos(\omega_d t+\angle k)$$

where

$$\omega_d = \omega_n\sqrt{1-\zeta^2}. \tag{6.3}$$

The following facts can be used to locate the roots in the complex plane:

a) the real part of the roots is equal to $-\zeta\,\omega_n$;

b) the absolute value of both roots is equal to

$$w_n\left|-\zeta \pm j\sqrt{1-\zeta^2}\right| = \omega_n\sqrt{\zeta^2+(1-\zeta^2)} = \omega_n;$$

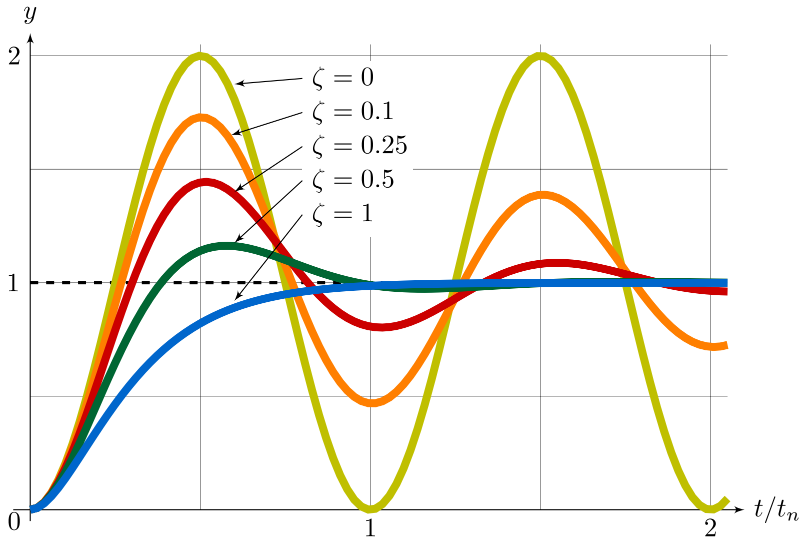


Fig. 6.2: Normalized step responses $y(t/t_n)$ for second order transfer-function $G(s) = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$ with $0 < \zeta < 1$, $t_n = 2\pi/\omega_n$

c) the angle measured between the root and the imaginary axis is

$$\phi_d = \sin^{-1}\frac{\zeta\omega_n}{\omega_n} = \sin^{-1}\zeta = \tan^{-1}\frac{\zeta}{\sqrt{1-\zeta^2}} = \tan^{-1}\frac{\zeta\,\omega_n}{\omega_d}. \tag{6.4}$$

These relationships can be visualized in the diagram of Fig. 6.1b.

The parameter ζ controls how damped the system response is, and it is know as the *damping ratio*. The name and the following terminology is borrowed from the standard analysis of the harmonic oscillator: when $\zeta = 0$ the the roots are purely imaginary and the response contains an oscillatory component at the *natural frequency* ω_n, i.e. $\omega_d = \omega_n$; when $0 < \zeta < 1$ the system is said to be *underdamped* as any response are oscillatory with *damped natural frequency* ω_d, $\omega_d < \omega_n$; when $\zeta = 1$ the system is said to be *critically damped*, with two repeated real roots; when $\zeta > 1$ the system has two real roots with negative real part and is said to be *overdamped*;

For a concrete example, let's take a look at the step response of a second-order system with characteristic equation (6.1):

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad \omega_n > 0. \tag{6.5}$$

The choice of ω_n^2 as the numerator is for normalization: the response of G to a unit step has steady-state gain $|G(0)| = 1$. The interesting case is when G is asymptotically stable but has complex-conjugate poles, that is $0 < \zeta < 1$.

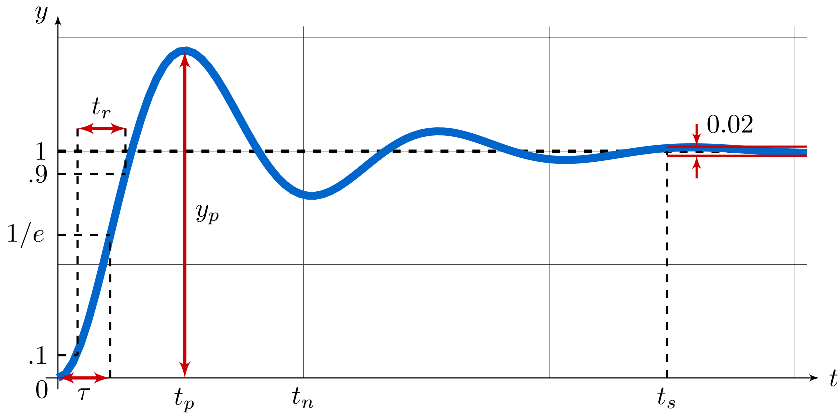


Fig. 6.3: Step response $y(t)$ showing peak time, t_p, settling-time, t_s, rise-time, t_r, peak response, y_p, and time-constant, τ

Using the methods of § 3.5, you will show in **P6.2** after a bit of algebra that the unit step response of a system with transfer-function (6.5) is:

$$y(t) = \mathcal{L}^{-1}\left\{\frac{G(s)}{s}\right\} = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin(\omega_d t + \pi/2 - \phi_d), \quad t \geq 0$$

where the damped natural frequency, ω_d, is given by (6.3), and ϕ_d, is given by (6.4). We plot the step response for various values of damping ratio, ζ, in Fig. 6.2. The plots are normalized by the parameter:

$$t_n = \frac{2\pi}{\omega_n}, \tag{6.6}$$

and the resulting function, $y(t/t_n)$, is a function of ζ only. The role of ω_n is therefore controlling the *speed* of the response: the larger the ω_n the faster the response. As for ζ, the larger the ζ the faster the rate of decay, that is, the *damping* of the response.

It is useful to define and quantify some terminology with respect to the step response of the standard second-order system (6.5) when $0 < \zeta < 1$. As seen in Fig. 6.2, there is always some *overshoot*, that is the response exceeds its steady-state value. This is a new dynamic characteristic, since strictly-proper first-order systems never overshoot. Differentiating the response with respect to t and looking for peaks we determine

$$t_p = \frac{\pi}{\omega_d}, \qquad y_p = 1 + e^{-\zeta\pi\big/\sqrt{1-\zeta^2}}, \tag{6.7}$$

which is the time the first peak occurs and its response value. It is common to quantify overshoot in terms of the *percentage overshoot*:

$$\mathrm{PO} = 100\, e^{-\zeta\pi \big/ \sqrt{1-\zeta^2}}. \tag{6.8}$$

Another useful metric is the *settling-time*, t_s, which is the time it takes the step response to become confined within 2 percent of its steady-state value. The settling-time can be approximated[2] by determining the time when

$$y(t) - 1 \approx \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\pi/2 - \phi_d) = e^{-\zeta\omega_n t} = \frac{1}{50},$$

which leads to

$$t_s = \frac{\log(50)}{\zeta\,\omega_n} \approx \frac{3.9}{\zeta\,\omega_n}.$$

These figures of merit are illustrated in Fig. 6.3. The *rise-time*, t_r, and the time-constant, τ, are also shown in this figure. They were computed in the exact same way as for first-order systems (see § 2.3): the rise-time is the time it takes for the response to go from 10% to 90% of its steady-state value, and the time-constant is the time it takes the response to reach $100\, e^{-1} \approx 63\%$ of its steady-state value. Unfortunately it is not possible to have simple formulas for the rise-time and the time-constant for second-order systems. The approximate formulas:

$$\frac{t_r}{t_n} \approx 0.16 + 0.14\,\zeta + 0.24\,\zeta^3, \qquad \frac{\tau}{t_n} \approx 0.19 + 0.1\,\zeta + 0.054\,\zeta^3$$

have a maximum relative error of about 1% in the range $0 < \zeta < 1$.

With the above discussion in mind, we revisit the cruise-control solution with the integral only control discussed in § 4.2. The closed-loop characteristic equation obtained in (4.11) can be put in the form (6.1) after setting

$$\omega_n^2 = \frac{p}{m} K, \qquad 2\,\zeta\,\omega_n = \frac{b}{m}.$$

In other words:

$$\omega_n = \sqrt{\frac{p}{m} K}, \qquad \zeta = \frac{b}{2\sqrt{p\,m\,K}} > 0.$$

From these relations it is clear that one cannot raise K in order to increase ω_n, i.e. the speed of the response, without compromising the damping ratio, ζ. By contrast, if proportional and integral control is used, the closed-loop poles are governed by the characteristic equation (4.16), which can be put in the form (6.1) with

$$\omega_n = \sqrt{\frac{p}{m} K_i}, \qquad \zeta = \frac{1}{2\,\omega_n} \left(\frac{b}{m} + \frac{p}{m} K_p \right) > 0.$$

[2] The approximation here is that $\omega_d t \approx 2k\pi$, $k \in \mathbb{Z}$.

One can now choose K_i to set the natural frequency, ω_n, to any desired value and then choose K_p to set the damping ratio, ζ, independently.

Consider as a second example the simple pendulum introduced in § 5.4. We will design and analyze sophisticated controllers for the simple pendulum in § 6.5 § 7.8, and § 8.1. But we can already take first steps based on our fresh analysis of second-order systems. The model derived for the simple pendulum in § 5.4 is nonlinear and we have obtained two linearized models around the equilibrium points $\theta = 0$ and $\theta = \pi$. The transfer-functions associated with these linearized models are:

$$G^0(s) = \frac{1}{J_r s^2 + bs + m\,g\,r}, \qquad G^\pi(s) = \frac{1}{J_r s^2 + bs - m\,g\,r},$$

where J_r is given by (5.14). As discussed in § 5.4, if all parameters are positive then the system linearized around $\theta = u = 0$ is asymptotically stable and the system linearized around $u = 0$, $\theta = \pi$, is unstable. Our first task is to develop a controller that can stabilize the linearized system G^π. Recalling the discussion in § 5.7, a linear controller that stabilizes a nonlinear system around its equilibrium point also stabilize the nonlinear system at least in a neighborhood of the equilibrium.

Consider the connection of the pendulum with the proportional controller $K(s) = K$, which can be represented by the block-diagram in Fig. 5.16. The feedback connection of the pendulum with this controller linearized in a neighborhood of $\theta = \pi$ has as closed-loop transfer-function[3]:

$$H^\pi(s) = \frac{K G^\pi(s)}{1 + K G^\pi(s)} = \frac{K}{J_r s^2 + bs + K - m\,g\,r}.$$

This is a second-order system with characteristic equation (6.1) where

$$\omega_n^\pi = \sqrt{\frac{K - m\,g\,r}{J_r}}, \qquad \zeta^\pi = \frac{b}{2\sqrt{J_r(K - m\,g\,r)}}.$$

These expressions are similar to the ones found in the case of the integral cruise control of the car, where increasing K improves the response speed of the response, ω_n, at the expense of decreasing the damping ratio, ζ. A complication is the fact that K needs to be chosen high enough to stabilize[1] the pendulum:

$$K > m\,g\,r.$$

It is desirable to have a controller that can work around both equilibrium points, that is near $\theta = 0$ and $\theta = \pi$. A controller which is capable of operating a system in a variety of conditions is a *robust* controller. We will discuss *robustness* in more detail later in § 7.7 and § 8.1. For now, we are happy to verify that the same proportional control does not de-stabilize the system when

[3] Work this out in detail in **P6.12**.

operated near the stable equilibrium $\theta = 0$. Repeating the same steps as above we obtain the closed-loop transfer-function:

$$H^0(s) = \frac{KG^0(s)}{1 + KG^0(s)} = \frac{K}{J_r s^2 + bs + K + m\,g\,r}$$

for which

$$\omega_n^0 = \sqrt{\frac{K + m\,g\,r}{J_r}}, \qquad \zeta^0 = \frac{b}{2\sqrt{J_r(K + m\,g\,r)}}.$$

We can see that a choice of gain, K, that stabilizes the unstable transfer-function G^π does not de-stabilize the transfer-function G^0 since $K + m\,g\,r > 2\,m\,g\,r > 0$. For example:

$$K = 3\,m\,g\,r, \tag{6.9}$$

implies

$$\omega_n^\pi = \sqrt{2}\,\omega_n^{\mathrm{ol}}, \qquad \omega_n^0 = 2\,\omega_n^{\mathrm{ol}}, \qquad \omega_n^{\mathrm{ol}} = \sqrt{\frac{m\,g\,r}{J_r}},$$

where ω_n^{ol} is the natural frequency of the stable transfer-function G^0 in open-loop, i.e. without control. The choice of gain (6.9) doubles the natural frequency near the stable equilibrium $\bar\theta = 0$ while stabilizing the unstable equilibrium $\bar\theta = \pi$ by making its closed-loop natural frequency be approximately 40% higher than ω_n^{ol}. As for the damping ratio, a simple calculation reveals

$$\zeta^0 = \frac{1}{2}\zeta^{\mathrm{ol}}, \qquad \zeta^\pi = \frac{1}{\sqrt{2}}\zeta^{\mathrm{ol}}, \qquad \zeta^{\mathrm{ol}} = \frac{b}{2\sqrt{J_r m\,g\,r}}.$$

The choice (6.9) halves the damping ratio near the stable equilibrium $\bar\theta = 0$ while providing a damping ratio of about 70% of the open-loop damping ratio, ζ^{ol}, around the unstable equilibrium $\bar\theta = \pi$. In both cases the controlled pendulum has a damping ratio which is smaller than the open-loop pendulum. The only way to improve the damping ratio is to reduce K, which makes the closed-loop system response slower. As seen in the car cruise control problem, a better solution was possible with a more complex controller: a proportional-integral controller enabled one to set the damping ratio and the natural frequency by a choice of two independent gains. As we will discuss in detail in the next section, the key in the case of the pendulum is the addition of a zero in the controller transfer-function: instead of an integrator we need a differentiator.

We close this section with a note on the behavior of some systems that have order higher than second. Even though it is not possible to provide detailed formulas for the response of such systems, an asymptotically stable higher-order system that has a real pole or a pair of complex conjugates poles with real part much smaller then the system other poles tends to behave like a first-

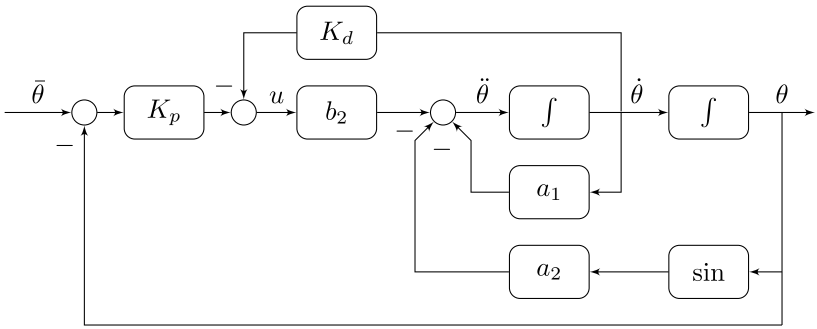


Fig. 6.4: Proportional-derivative control of the pendulum

or a second-order system. Indeed, the response becomes quickly *dominated* by the slower response of such poles: the contribution of faster and highly damped poles approaches zero quicker and leave the response appearing to be that of a first- or second-order system, the slow poles being called *dominant* poles. For this reason, it is common to see open- and closed-loop system's performance specifications in terms of first- or second-order figures of merit.

6.2 Derivative Action

We have seen in previous sections that increasing the control gain may have a detrimental effect on the damping properties of a feedback system. This has been the case in the integral cruise control as well as in the proportional control of the pendulum. In order to improve damping we generally need to increase the complexity of the controller. In a mechanical system, the idea of damping is naturally associated with a force that opposes and increases with the system velocity, for instance, viscous friction in the simple pendulum model, $-b\,\dot{\theta}$. Using velocity or, more generally, the derivative of a signal, is an effective way to improve damping. However, as it will become clear later in § 6.4, velocity feedback alone is usually not enough to stabilize an unstable system, and hence derivative action often needs to be combined with proportional control.

We introduce the idea of proportional-derivative control using the simple pendulum as an example. Let the proportional-derivative controller

$$u(t) = K_p\left(\bar{\theta}(t) - \theta(t)\right) - K_d\,\dot{\theta}(t) \tag{6.10}$$

be applied to the simple pendulum model derived in § 5.4. This controller can be implemented as shown in the block-diagram of Fig. 6.4, in which the controller makes use of two measurements: the angular position, θ, and the angular velocity, $\dot{\theta}$. If we are not able to directly measure the velocity, $\dot{\theta}$, then we have to be careful when building the controller, given the difficulties to physically implement a derivative block, as discussed in § 5. Either way, in order to avoid

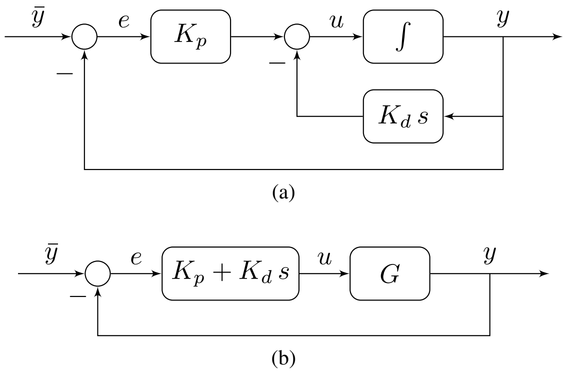


Fig. 6.5: Block-diagrams for proportional-derivative control

working with two outputs, it is convenient to introduce a derivative block component during controller analysis and design. With this caveat in mind, we can study the single output block diagrams in Fig. 6.5, where we use the Laplace variable 's' to represent an idealized derivative operation[4].

The closed-loop systems in Fig. 6.5a and Fig. 6.5b are not equivalent, and behave slightly different when $\bar{y}$ is not constant. The diagram in Fig. 6.5a corresponds to Fig. 6.4, which implements the control law (6.10). The diagram in Fig. 6.5b, is the standard *proportional-derivative* controller, or PD controller:

$$u(t) = K_p\, e(t) + K_d\, \dot{e}(t), \qquad e(t) = \bar{y}(t) - y(t). \tag{6.11}$$

From an implementation perspective, the diagram in Fig. 6.5a might offer some advantages, specially when $\bar{y}$ has discontinuities or is rich in high-frequency components. Note that y is continuous even if $\bar{y}$ is not. For this reason, in the presence of discontinuities, one should expect to see short-lived large signal spikes[5] in the control input u in Fig. 6.5b when compared to Fig. 6.5a. This is due to the differentiation of $\bar{y}$ in Fig. 6.5b. In terms of transfer-functions, this difference amounts to an extra zero, with both diagrams having the same closed-loop poles. To see this let $G = N_L/D_L$, and verify that

$$H(s) = \frac{K_p \frac{N_L(s)}{D_L(s)+K_d s N_L(s)}}{1 + K_p \frac{N_L(s)}{D_L(s)+K_d s N_L(s)}} = \frac{K_p N_L(s)}{D_L(s) + (K_p + K_d s) N_L(s)}$$

for the loop in Fig. 6.5a whereas

$$H(s) = \frac{(K_p + K_d s)\frac{N_L(s)}{D_L(s)}}{1 + (K_p + sK_d)\frac{N_L(s)}{D_L(s)}} = \frac{(K_p + K_d s) N_L(s)}{D_L(s) + (K_p + K_d s) N_L(s)}$$

[4] Recall from Tab. 3.2 that $\mathcal{L}(\dot{f}(t)) = sF(s) - f(0)$.

[5] See the discussion in § 5.1 about impulses appearing in the response.

in Fig. 6.5b. Since Fig. 6.5b is a better fit for the feedback diagram of Fig. 4.8 which we have analyzed earlier, we shall use Fig. 6.5b in the rest of this section.

Consider the control of the simple pendulum by the proportional-derivative controller in Fig. 6.5b. Linearizing about $u = 0$ and $\theta = \pi$ we obtain the linearized closed-loop transfer-function (see **P6.13**):

$$H^{\pi}(s) = \frac{(K_p + K_d s)G^{\pi}(s)}{1 + (K_p + sK_d)G^{\pi}(s)} = \frac{K_p + K_d s}{J_r s^2 + (K_d + b)s + (K_p - m\,g\,r)}$$

Because the controller does not have any poles, this is also a second-order system with characteristic equation (6.1) for which

$$\omega_n^{\pi} = \sqrt{\frac{K_p - m\,g\,r}{J_r}}, \qquad \zeta^{\pi} = \frac{K_d + b}{2\sqrt{J_r(K_p - m\,g\,r)}}.$$

The same calculations repeated for $\bar{\theta} = 0$ produce

$$\omega_n^{0} = \sqrt{\frac{K_p + m\,g\,r}{J_r}}, \qquad \zeta^{0} = \frac{K_d + b}{2\sqrt{J_r(K_p + m\,g\,r)}}.$$

Note how the values of ω_n are not affected by the introduction of the derivative gain, K_d, which leads to the important conclusion that damping alone (derivative action), would not have been able to stabilize the pendulum. The effect of K_d is confined to the damping ratio, ζ. For example, after stabilizing the closed-loop pendulum with the choice of $K_p = 3\,m\,g\,r$ from (6.9) we can choose K_d to provide any desired level of damping, say

$$\zeta = \zeta^{\pi} = \frac{\sqrt{2}}{2} \approx 0.7$$

around the equilibrium $\bar{\theta} = \pi$. This corresponds to setting

$$K_d = 2\,\zeta^{\pi}\sqrt{J_r(K_p - m\,g\,r)} - b = 2\sqrt{J_r m\,g\,r} - b.$$

For such K_p and K_d, the damping ratio around the equilibrium $\bar{\theta} = 0$ is

$$\zeta^{0} = \frac{2\sqrt{J_r m\,g\,r}}{2\sqrt{4 J_r m\,g\,r}} = \frac{1}{2},$$

which is a bit smaller than ζ^{π}, but still acceptable. We will continue to improve this design in § 6.5 and § 7.8.

6.3 Proportional-Integral-Derivative Control

Derivative action can help improve the damping properties of a feedback system, as seen in the last section. If integral action is required to track constant or

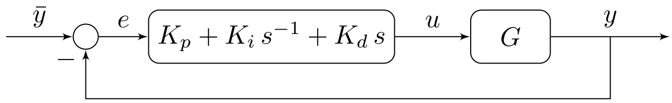


Fig. 6.6: Block-diagram for Proportional-Integral-Derivative (PID) control

low-frequency references, or reject low-frequency input disturbances, a controller can be constructed combining a proportional term with integral *and* derivative terms to form a *proportional-integral-derivative* controller, or PID controller. A generic block-diagram of a feedback system with a PID controller is shown in Fig. 6.6. PID control is arguably the most popular form of control and is widely used in industry. Manufacturers of instrumentation and control hardware offer implementations of PID controllers where the gains K_p, K_i and K_d can be tuned by the user to fit a specific controlled process. Most modern hardware come with algorithms for identifying the process and automatically tuning the gains, and other bells and whistles.

The PID controller in Fig. 6.6 implements the control law

$$u(t) = K_p\, e(t) + K_d\, \dot{e}(t) + K_i \int_0^t e(\tau)\, d\tau, \quad e(t) = \bar{y}(t) - y(t), \tag{6.12}$$

which is associated with the transfer-function

$$K(s) = K_p + K_d s + \frac{K_i}{s} = \frac{K_d s^2 + K_p s + K_i}{s}. \tag{6.13}$$

This transfer-function has two zeros and one pole at the origin. When the derivative gain K_d is not zero, this transfer-function is not proper and cannot be physically implemented, an issue which we will address later in § 6.5. Given the importance and widespread use of PID control, there is an extensive literature on how to select the control gains based on assumptions on the underlying model of the system being controlled. We do not study such rules here, as they can be found in standard references, e.g. [FPEN09, DB97]. These *tuning rules* might many times provide a useful and quick way to have a feedback system up and running based on a few parameters that can be obtained experimentally, but do little to elucidate the role of the controller and its relation with the controlled system in a feedback loop. For this reason we shall spend our time studying methods that can help guide the selection of the PID control gains but require a more complete model of the system. One difficulty is that the controller (6.13) is of order one, and application to a simple second-order system generally leads to a third-order closed-loop system. The situation gets even more complicated if additional poles are introduced to make (6.13) proper and therefore implementable, in which case the order of the PID controller is at least two. Calculating the roots of a closed-loop characteristic polynomial of order three or four in terms of the parameters K_p, K_i and K_d is possible but requires working with highly convoluted formulas. Let alone that it is impossible to come up

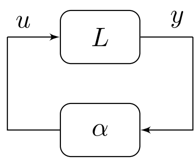


Fig. 6.7: Closed-loop feedback configuration for root-locus

with any type of analytic formula for computing the roots of a polynomial of order five or higher except in very special cases[6]. Instead, we should rely on numerical computations and auxiliary graphic tools which we study next and in the following chapters.

6.4 Root-Locus

In this section we introduce a technique for controller design based on a graphic know as *root-locus*. The root-locus is a plot showing the poles of a system with transfer-function L in feedback with a static gain, α, as shown in Fig. 6.7, that is a plot of the roots of the characteristic equation:

$$1 + \alpha\, L(s) = 0, \qquad\qquad \alpha \geq 0,$$

obtained as α varies from 0 to infinity. The root-locus plot is used to study general SISO feedback systems after properly grouping all dynamic elements into the *loop transfer-function*, L.

Take for instance the diagram of Fig. 4.14. As shown in § 4, the closed-loop poles are the zeros of the characteristic equation

$$1 + G(s)\, K(s)\, F(s) = 0.$$

The transfer-function $G(s)K(s)F(s)$ is the product of all transfer-functions appearing in the feedback loop. Now let the controller $K(s)$ be divided in two parts: a *positive* static gain α and a dynamic part $C(s)$, that is

$$K(s) = \alpha\, C(s), \qquad \alpha > 0.$$

In this way, the characteristic equation is rewritten as

$$1 + \alpha\, L(s) = 0, \qquad\qquad L(s) = G(s)\, C(s)\, F(s),$$

and the root-locus plot is used for the determination of a suitable gain α.

As mentioned earlier, one reason for working with a graphic rather than an algebraic tool is that it is not possible to find a closed-form expression for the roots of a polynomial of arbitrary degree as a function of its coefficients, even

[6]This is the Abel-Ruffini Theorem.

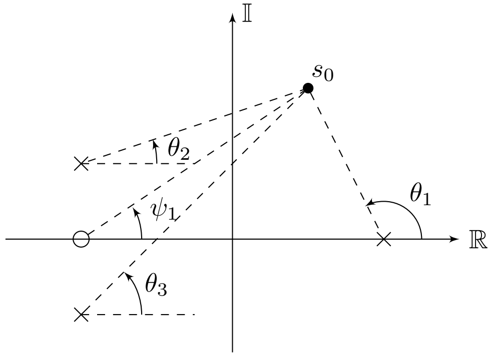


Fig. 6.8: Measuring the phase of $L(s)$ at $s = s_0$; '$\times$' denotes a pole of $L(s)$ and '$\circ$' denotes a zero of $L(s)$; $\angle L(s_0) = \psi_1 - \theta_1 - \theta_2 - \theta_3$; the point s_0 shown is not a root of $1 + \alpha\, L(s)$ because $\angle L(s_0) \neq \pi$

in this simple case where the roots depend on only one parameter, α. It is however relatively simple to *sketch* the location of the roots with respect to α without computing their exact location. This can even be done *by hand* based on a set of *rules* that depend only on the computation of the zeros and poles of $L(s)$. With the help of a computer program, such as MATLAB, very accurate root-locus plots can be traced without effort. Knowledge of some but not necessarily all of the root-locus rules can help one predict the effect of moving or adding extra poles or zeros to L, which is a very useful skill to have when designing controllers and analyzing feedback systems. It is with this intent that we introduce some of the root-locus rules. Readers are referred to standard references for a complete set of rules [FPEN09, DB97].

A key observation behind the root-locus is the following property: suppose the complex number s is a root of $1 + \alpha L(s)$ and $\alpha > 0$. In this case:

$$L(s) = -\frac{1}{\alpha} < 0 \qquad \Longrightarrow \qquad \angle L(s) = \pi. \tag{6.14}$$

In other words, $L(s)$ is a negative real number and hence with phase equal to π. Conversely, suppose that $\angle L(s) = \pi$ for some s, then $L(s)$ is a negative real number, that is $L(s) = -\alpha^{-1}$ for some $\alpha > 0$ and $1 + \alpha L(s) = 0$. It is much easier to locates points in the complex plane for which $L(s)$ has phase equal to π than to compute the roots of $1 + \alpha\, L(s)$ as a function of α. This is specially simple when $L(s)$ is rational, as discussed below.

Let $L = N_L/D_L$ be a rational transfer-function where n is the degree of the denominator, D_L, and m the degree of the numerator, N_L. Factor N_L and D_L into the product of their possibly complex roots:

$$\begin{aligned} N_L &= (s - z_1)(s - z_2) \cdots (s - z_m), \\ D_L &= (s - p_1)(s - p_2) \cdots (s - p_n). \end{aligned}$$

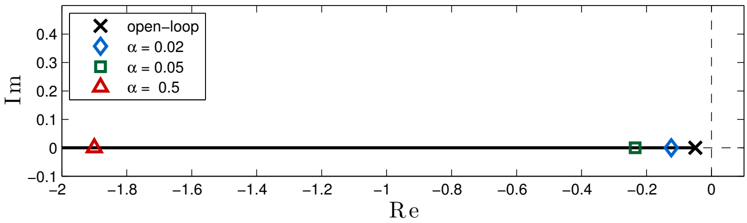


Fig. 6.9: Root-locus for the proportional control of the car speed model (2.3) with parameters $b/m = 0.05$ and $p/b = 73.3$; diamonds show roots for the gains indicated in the legend; compare with the step responses in Fig. 2.10

Using the fact that

$$\angle L = \angle N_L - \angle D_L = \sum_{i=1}^{m} \angle(s - z_i) - \sum_{k=1}^{n} \angle(s - p_k)$$

the phase of L, i.e. $\angle L$, can be quickly computed by evaluating angles measured from the zeros $z_1, z_2, \cdots, z_m$ and poles $p_1, p_2, \cdots, p_n$. This is the key idea behind sketching the root-locus plot and is illustrated in Fig. 6.8.

Because L and $\angle L$ are continuous function of s at any point that is not a pole of L, that is a root of D_L, one should expect that the root-locus be a set of *continuous curves* in the complex plane. Indeed, this fact and more is part of our first *root-locus rule*:

> If the rational function $L = N_L/D_L$ is strictly proper, $n \geq m$, and there is no pole-zero cancelation between N_L and D_L, the characteristic equation $1 + \alpha\, L(s) = 0$ has exactly n roots for any $\alpha > 0$. These roots belong to n continuous curves in the complex plane (the root-locus). When $\alpha \to 0$ these roots are the roots of the denominator of D_L, and when $\alpha \to \infty$ these roots are the roots of the numerator of N_L plus $n - m$ roots at infinity.

The roots *depart* from the poles and *arrive* at the zeros of L because

$$\lim_{\alpha \to 0} D_L(s)\left(1 + \alpha \frac{N_L(s)}{D_L(s)}\right) = \lim_{\alpha \to 0} D_L(s) + \alpha N_L(s) = D_L(s),$$

$$\lim_{\alpha \to \infty} \alpha^{-1} D_L(s)\left(1 + \alpha \frac{N_L(s)}{D_L(s)}\right) = \lim_{\alpha \to \infty} \alpha^{-1} D_L(s) + N_L(s) = N_L(s),$$

for any s such that $D_L(s) \neq 0$, that is any s that is not a pole of L.

Take for example the linear model we have obtained for the car in (2.3) and the parameters $b/m = 0.05$ and $p/b = 73.3$ estimated in § 2. This model has an open-loop transfer-function (3.9):

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}.$$

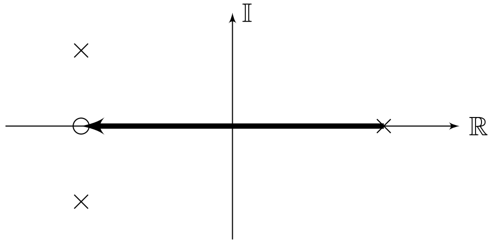


Fig. 6.10: Locus of real roots

The stability of the feedback look with proportional control in Fig. 2.7 can be studied with the root-locus plot after setting

$$\alpha = K, \qquad L = G = \frac{N_L}{R}, \qquad N_L = 3.7, \qquad D_L = s + 0.05,$$

The root-locus plot in this case is very simple, the black solid segment of line in Fig. 6.9. Because $n = 1$, $m = 0$, it consists of a single curve starting at the open-loop root of D_L, marked with a cross, and ending at the $n - m = 1$ zero of N_L, which is at minus infinity. The markers in Fig. 6.9 locate selected values of α that correspond to the control gains used to generate Fig. 2.10. There are no other curves in this root locus. It is not possible for the closed-loop system to have complex roots because L is of order one, and the root-locus remains entirely on the real axis.

In general, when L is a rational function with real coefficients, it is possible to quickly determine which parts of the real axis belong to the root locus. Consider first the case where L has only real poles and real zeros. If s is a point on the real axis, then zeros and poles to the left of s contribute nothing to the phase of L, whereas zeros and poles to the right of s contribute either π or $-\pi$. Therefore, whenever there is an odd number of real poles or real zeros to the right of the real point s, this point should be part of the root-locus. If there is an even number of real poles and real zeros to the right of the real point s, this point shoud not be part of the root-locus. After observing that complex zeros or poles always come in complex-conjugate pairs and that the net contribution of a complex-conjugate pair of zeros or poles to the phase of L for a point s on the real axis is always 0 if the zero or pole is to the left of s or $\pm 2\pi$ if the zero or pole is to the right[7] of s, we can come up with the rule:

> The real point s is part of the root-locus of the rational function $L = N_L/D_L$ where N_L and D_L are polynomials with real coefficients if and only if L has an odd number of poles and zeros with real part greater than or equal to s.

This rule is all that is needed to explain the root-locus of Fig. 6.9. For another example, take the poles and zeros in Fig. 6.8. Application of the rule leads to

[7] Imagine what happens if the point s_0 is placed on the real axis in Fig. 6.8.

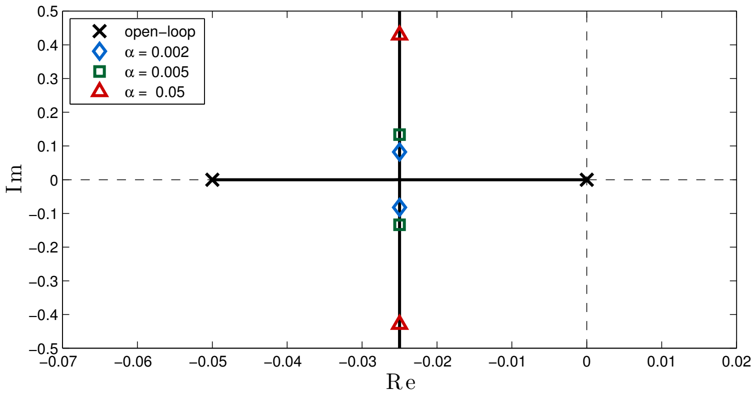


Fig. 6.11: Root-locus for the integral control of the car speed model (2.3) with parameters $b/m = 0.05$ and $p/b = 73.3$; diamonds show roots for the gains indicated in the legend; compare with the step responses in Fig. 4.4

the real segment shown in Fig. 6.10 being the only part of the root locus on the real axis. The arrow indicates the fact that the locus depart from the pole and arrive at the zero. We will add the remaining curves to this root-locus later.

When L is of order 2 or higher the root-locus will most likely have complex roots. Consider for example the car model

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05},$$

this time in closed-loop with the integral controller shown in Fig. 4.3. In this case, $K(s) = KC(s)$, $C(s) = s^{-1}$ and the root-locus computed with

$$\alpha = K, \qquad L = GC = \frac{N_L}{D_L}, \qquad N_L = 3.7, \qquad D_L = s(s + 0.05),$$

is shown in Fig. 6.11. In this example $n = 2$, $m = 0$, and we have two curves (solid segments of line) starting at the roots of D_L, $s = 0$ and $s = -0.05$, and ending at the $n - m = 2$ zeros of N_L, which are at infinity. The two curves meet at the *break-away point* $s = -0.025$, where the two roots become complex-conjugate pairs. The markers locate values of α that correspond to the control gains used to generate Fig. 4.4. This figure helps us understand why this example's transient performance with integral only control is much worse than with proportional only control: in the I case, the closed-loop roots, can never have real part more negative than the open-loop pole $s = -0.05$, whereas in the P case, the closed-loop roots always have real part more negative than the open-loop pole. Compare Fig. 6.11 with Fig. 6.9. The behavior after the roots break-away from the real line is explained by the next rule.

If the rational function $L = N_L/D_L$ where N_L and D_L are polynomials with real coefficients is stricly proper, $n > m$, then there are $n-m$ roots that converge to a set of $n-m$ straight line asymptotes which intesect at the real point

$$c = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{n} z_i}{n-m}.$$

The angle that the asymptotes make with the real axis is

$$\phi_i = \frac{\pi + i2\pi}{n-m}, \quad i = 0, \cdots, n-m-1.$$

In Fig. 6.11 we have $n - m = 2$ and two straight line asymptotes intersecting

$$c = \frac{-0.05 + 0}{2} = -0.025$$

at angles

$$\phi_1 = \frac{\pi}{2}, \qquad\qquad \phi_2 = \frac{3\pi}{2}.$$

The behavior out of the real axis in Fig. 6.11 is completely determined by the asymptotes. On a more complicated example, the finite locuss depend on the location of the poles and zeros. For the pole-zero configuration of Fig. 6.8 we have $n = 3$, $m = 1$, hence $n - m = 2$ asymptotes. The exact values of the poles and zeros are not given in the figure, but we can assume by symmetry that

$$p_1 = a, \qquad p_2 = -a + jb, \qquad p_2 = -a - jb, \qquad z_1 = -a,$$

for some real $a > 0$ and $b > 0$. Hence, the asymptotes should intersect at

$$c = \frac{(-a - jb) + (-a + jb) + (a) - (-a)}{3 - 1} = 0$$

which is the origin. That is, the two asymptotes coincide with the imaginary axis. The path followed by the complex root-locus will however depend on the particular values of a and b. Fig. 6.12 illustrates three possibilities: in Fig. 6.12a the complex locus never intersects the real locus; in Fig. 6.12b the complex locus and the real locus merge at break-away points where the roots have multiplicity two; finally in Fig. 6.12c the complex locus and the real locus touch at a single break-away point where the roots have multiplicity three. Which one will take place depends on the particular values of the open-loop poles and zeros. There are rules to determine the existence of multiple roots on the real axis which allow one to distinguish these three cases. These days, however, the best practical way to determine which case occurs is with the help of a computer software, such as MATLAB. Note that, intuitively, the root paths tend to *attract* each other. So if the imaginary component of the complex conjugate

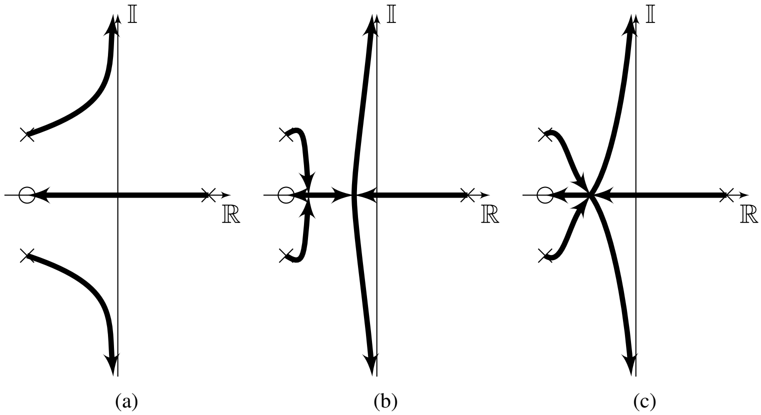


Fig. 6.12: Possible root-locus for transfer-function with three poles and one zero

poles is large, Fig. 6.12a is likely to occur, $b/a = 1$; whereas if the imaginary component is small Fig. 6.12b is likely to occur, $b/a = 1/4$; finally Fig. 6.12c represents the transition from Fig. 6.12a to Fig. 6.12b, and occurs only for very special values of a and b, in this example only if $b/a = 2\sqrt{3}/9 \approx 0.38$.

Finally the car cuise control model with the proportional and integral PI controller in Fig. 4.5 has

$$G = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}, \qquad K(s) = K_p + \frac{K_i}{s}$$

We let $K(s) = K_p C(s)$, $C(s) = s^{-1}(s + K_i/K_p)$, and

$$\alpha = K_p, \quad L = GC = \frac{N_L}{D_L}, \quad N_L = 3.7\left(s + \frac{K_i}{K_p}\right), \quad D_L = s\,(s + 0.05),$$

and compute the root-locus after setting

$$\frac{K_i}{K_p} = \gamma \frac{b}{m} = \gamma\, 0.05$$

for various values of γ. The pole-zero cancelation choice of (4.17) corresponds to $\gamma = 1$. With cancelation, the order of the closed-loop system is reduced to first-order, and the root-locus is a straight line, as can be seen in the third plot in Fig. 6.13 (the canceled pair of pole and zero is still displayed). Other plots in Fig. 6.13 show the root-locus when the cancelation is not exact, that is $\gamma \neq 1$. When we overestimate the value of the pole the controller zero is placed on the left of the pole and the root-locus must branch out of the real axis to reach the zeros, one of which is at infinity. When we understimate the value of the pole the controller zero is placed on the right of the pole and the root-locus has a

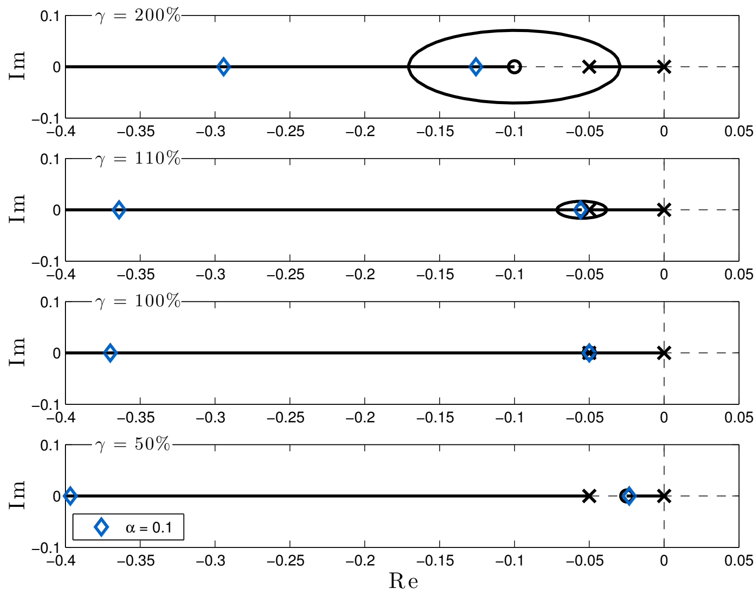


Fig. 6.13: Root-locus for the proportional-integral control of the car speed model (2.3) with parameters $b/m = 0.05$ and $p/b = 73.3$; diamonds show roots for a proportional gains of $K_p = 0.1$ and integral gain $K_i/K_p = \gamma\, b/m$; when $\gamma = 1$ (100%) the controller performs an exact pole-zero cancelation; compare with the step responses in Fig. 4.15

root which is necessarily slower than the value of this zero. The choice of roots marked with diamonds correspond to controllers used to plot the step responses in Fig. 4.15.

6.5 Control of the Simple Pendulum - Part I

We will now work a complete design example: the control of the simple pendulum. The goal is to design a controller that can drive the pendulum to any desired angular position. That is, we want to track a given reference angle around both stable and unstable equilibrium points. We start as we left off in § 6.2, where we designed a proportional-derivative (PD) controller with transfer-function:

$$C_{\mathrm{PD}}(s) = K_d\,(s + z)\,, \qquad z = \frac{K_p}{K_d},$$

that was able to stabilize the simple pendulum around both equilibrium points. The choice of gains

$$K_p = 3\,m\,g\,r, \qquad K_d = 2\sqrt{J_r m\,g\,r} - b$$

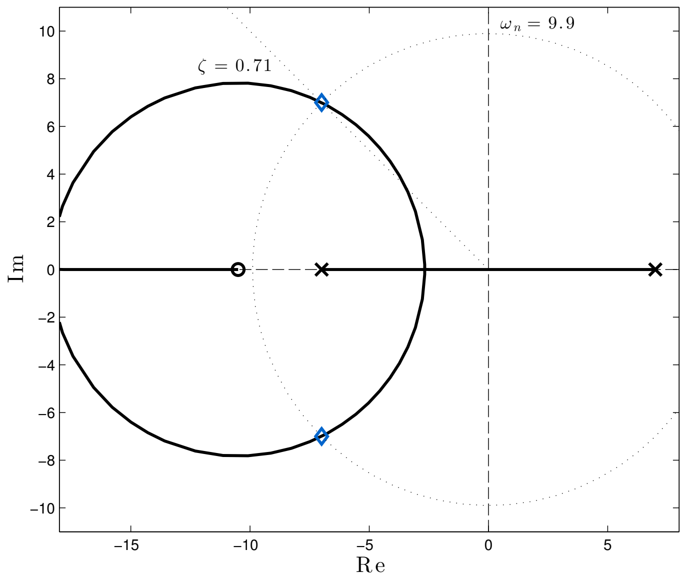


Fig. 6.14: Root-locus for the proportional-derivative control of the simple pendulum; diamonds mark the choice of gains from § 6.2

was shown to produce a stable second-order closed-loop system around the unstable equilibrium $\bar{\theta} = \pi$ with natural frequency $\omega_n^\pi = \sqrt{2}\,\omega_n^{\text{ol}}$ and a damping factor $\zeta^\pi = \sqrt{2}/2$. In order to better quantify the performance of this controller we consider the following numerical parameters:

$$m = 0.5 \text{ Kg}, \quad \ell = 0.3 \text{ m}, \qquad r = \ell/2 = 0.15 \text{ m}, \tag{6.15}$$

$$b = 0 \text{ Kg/s}, \quad g = 9.8 \text{ m/s}^2, \quad J = \frac{m\ell^2}{12} = 3.75 \times 10^{-3} \text{ Kg m}^2. \tag{6.16}$$

Note that we are designing our controller based on a friction-less pendulum model, $b = 0$, so that all damping must be provided by the controller. We compute the open-loop natural frequency around the stable equilibrium $\bar{\theta} = 0$:

$$\omega_n^{\text{ol}} = 7$$

and the controller

$$K_d \approx 0.21, \qquad K_p \approx 2.21, \qquad z \approx 10.5, \qquad C_{\text{PD}}(s) \approx 0.21(s+z).$$

A root-locus diagram corresponding to this controller is obtained with:

$$\alpha = K_d, \qquad L = \frac{1}{K_d} G^\pi\, C_{\text{PD}} = \frac{N_L}{D_L}, \qquad N_L = (s+z), \qquad D_L = 1,$$

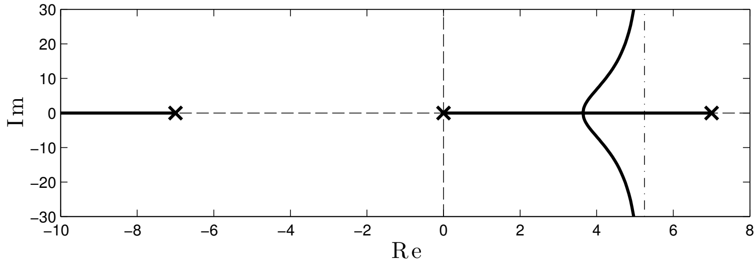


Fig. 6.15: Root-locus for the proportional-derivative control of the simple pendulum when the controller has an additional pole at the origin; no positive choice of gain makes the closed-loop stable

in Fig. 6.14. Diamond markers corresponding to $\alpha = K_d \approx 0.21$ confirm the predicted closed-loop natural frequency and damping factor:

$$\omega_n^\pi = \sqrt{2}\,\omega_n^{\mathrm{ol}} \approx 9.90, \qquad \zeta^\pi = \frac{1}{\sqrt{2}} \approx 0.71.$$

As commented in § 6.2, in order to physically implement this PD controller we need to have a direct measurement of the pendulum angular velocity, since the controller transfer-function is not proper. If this is not an option, then we need an alternative strategy. Recalling our ultimate goal of achieving arbitrary angular tracking one might be tempted to simply add a pole at zero to the controller, obtaining a proper controller of the form:

$$C_{\mathrm{PD+I}}(s) = \alpha \frac{s+z}{s}, \qquad z > 0.$$

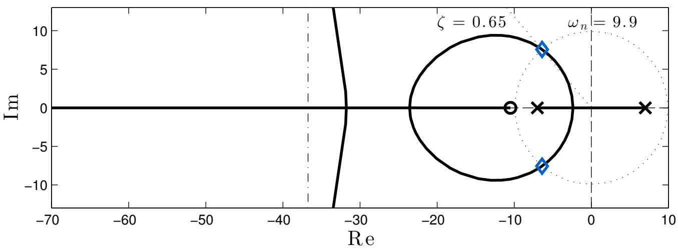
The problem with the introduction of the extra pole at the origin is that this controller no longer stabilizes the closed-loop system. Indeed, the root-locus now has a path between the right-most pole of the pendulum and the origin which must *escape* the real axis in order for the closed-loop system to become stable. The introduction of the pole means that the locus has $n-m = 3-1 = 2$ asymptotes centered at

$$c = \frac{\sqrt{m\,g\,r} - \sqrt{m\,g\,r} + 0 - (-z)}{n-m} = \frac{z}{2} > 0$$

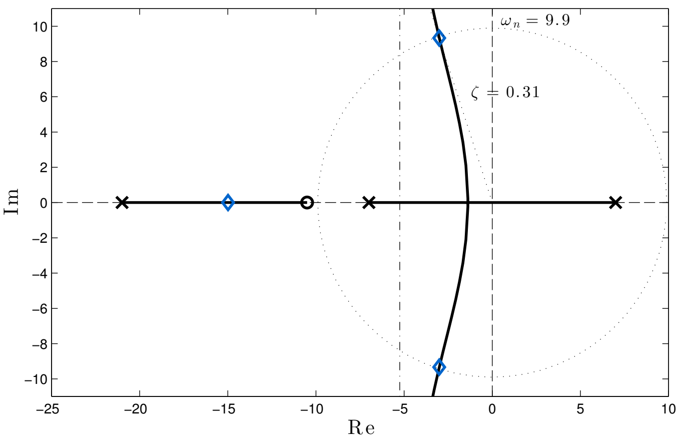
and most likely the roots on the right-hand side of the complex plane will simply converge toward the two vertical asymptotes which have positive real part. This is illustrated by the root-locus in Fig. 6.15.

A conclusion from the analysis of the asymptotes is that if a pole, $-p$, is to be added to make $C_{\mathrm{PD}}(s)$ a proper controller it should be added to the left of the zero, $-z$. In this way $p > z$ and the pair of asymptotes is now centered at

$$c = \frac{\sqrt{m\,g\,r} - \sqrt{m\,g\,r} + (-p) - (-z)}{n-m} = \frac{z-p}{2} < 0,$$



(a) Pole at $p = 8z$



(b) Pole $p = 2z$

Fig. 6.16: Root-locus for the lead control of the simple pendulum; when the pole is placed far enough on the left of the zero (a) the root-locus near the origin is similar to Fig. 6.14; as the pole is brought closer to the zero (b) the root-locus behavior changes; for the same closed-loop natural frequency, $\omega_n \approx 9.98$, the closed-loop damping ratio, $\zeta \approx 0.31$, achievable for $p = 2z$ is much smaller

on the left-hand side of the complex plane. This results in the controller:

$$C_{\text{lead}}(s) = K\frac{s+z}{s+p}, \qquad p > z > 0, \qquad K > 0.$$

For reasons which will be clear in § 7, such controller is known as a *lead compensator*. We use the root-locus diagram to help us select the position of the pole, that is the value of p. This is accomplished by defining the loop transfer-function:

$$\alpha = \frac{K}{pK_d}, \quad L = G^{\pi} C_{\text{lead}}, \quad C_{\text{lead}}(s) = p\,K_d \frac{s+z}{s+p}, \quad p > z = \frac{K_p}{K_d} > 0.$$

Note that

$$C_{\text{lead}}(s) \approx K_p + sK_d = C_{\text{PD}}(s) \qquad \text{when } s \approx 0$$

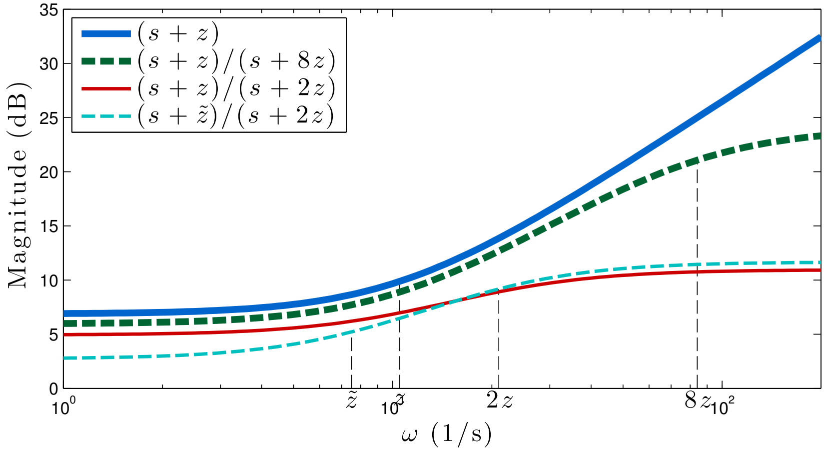


Fig. 6.17: Magnitude of the frequency response of the proportional-derivative controller and lead controllers with different choices for the pole and zero

which will help preserve the low-frequency behavior of the PD controller designed before. In this sense, we expect to select a root-locus gain α close to one. Moreover

$$C_{\text{lead}}(s) \approx K_p + sK_d = C_{\text{PD}}(s) \qquad \text{when } p \text{ is large.}$$

Hence, if we locate the pole $s = -p$ far enough on the left of the zero, we should expect that the new pole will not disturb the behavior of the closed-loop system. Indeed, this is what we see when placing the pole 8 times to the left of z, i.e. $p = 8z$, on the root-locus in Fig. 6.16a. Note that the region close to the origin is essentially the same as in Fig. 6.14, with the new asymptotes staying on the far left side of the plot. It is still possible to select a gain, α, that places the dominant closed-loop poles with a natural frequency, ω_n, and damping ratio, ζ, which are very close to the ones obtained with the PD controller. This choice of gain is marked with diamonds.

Placing the pole, $-p$, too far to the left means that the controller still has to deliver high gains in a wide region of the frequency spectrum. We compare the Bode plot the magnitude of the frequency response of the PD controller, that is $20\log_{10}|C_{\text{PD}}(j\omega)|$ (dB) versus $\log_{10}\omega$, in Fig. 6.17. We will be much better equipped to understand such plots in § 7. For now we just want to compare the magnitude of the frequency response of the PD controller, $(s+z)$ in Fig. 6.17, with the lead controller with the pole at $p = 8z$, $(s+z)/(s+8z)$ in Fig. 6.17. The lead controller still provides large amplification at high frequencies: of the order of 10 times higher (20 dB, see § 7.1) at $\omega = 8z$ than at $\omega = z$. With that in mind, we will attempt to bring the pole *closer* to the zero.

Placing the pole two times to the left of z, that is $p = 2z$, the resulting controller has to deliver much less gain at high frequencies, about 5dB or 1.8

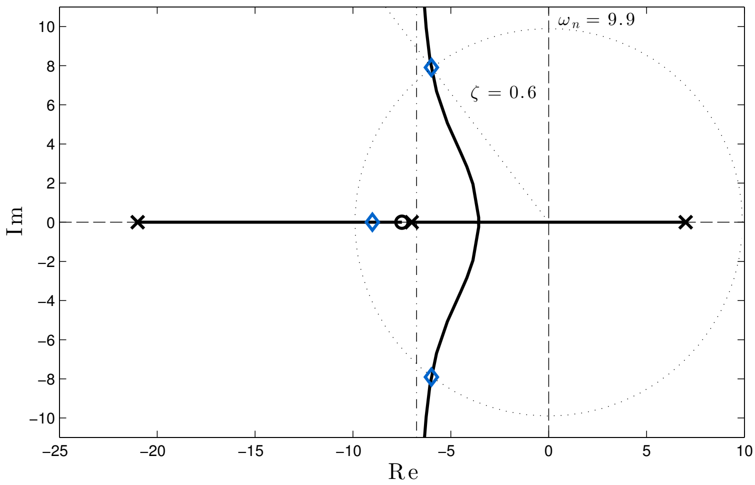


Fig. 6.18: Root-locus for the lead control of the simple pendulum after fixing the pole and adjusting the zero in order to improve the closed-loop damping ratio, $\zeta \approx 0.6$

times higher than at low-frequencies, as seen for $(s + z)/(s + 2z)$ in Fig. 6.17. However, the closed-loop behavior is much impacted, with the root-locus taking a completely different shape, Fig. 6.16b. The asymptotes are now located between the two system poles which results in much less damping, $\zeta \approx 0.31$, if ω_n is to stay close to $\omega_n \approx 9.90$. A comparable damping ratio is now only possible if the closed-loop system operates at a much slower natural frequency, which is not what we want in this case.

In order to improve the closed-loop damping ratio we fix the pole at $p = 2z$ and shift the zero of the controller toward the right in an attempt to move the asymptotes more to the left. This forces the intersection of the locus with the circle of $\omega_n = 9.90$ at higher damping ratios. After moving the zero to

$$\tilde{z} \approx 7.5$$

we obtain the root-locus shown in Fig. 6.18. It is now possible to select

$$K \approx 3.83$$

such that $\omega_n \approx 9.90$ and still obtain a damping ratio of about 0.6, which is a huge improvement when compared with the 0.31 damping-ratio obtained with the zero in its original location. As a side effect, as seen for $(s + \tilde{z})/(s + 2z)$ in Fig. 6.17, the resulting controller operates with lower gains at lower frequencies. Interestingly, after moving the zero we almost performed a pole-zero cancellation of the stable pole of the pendulum! The resulting controller is

$$C_{\text{lead}}(s) \approx 3.83 \frac{s + 7.5}{s + 21}. \tag{6.17}$$

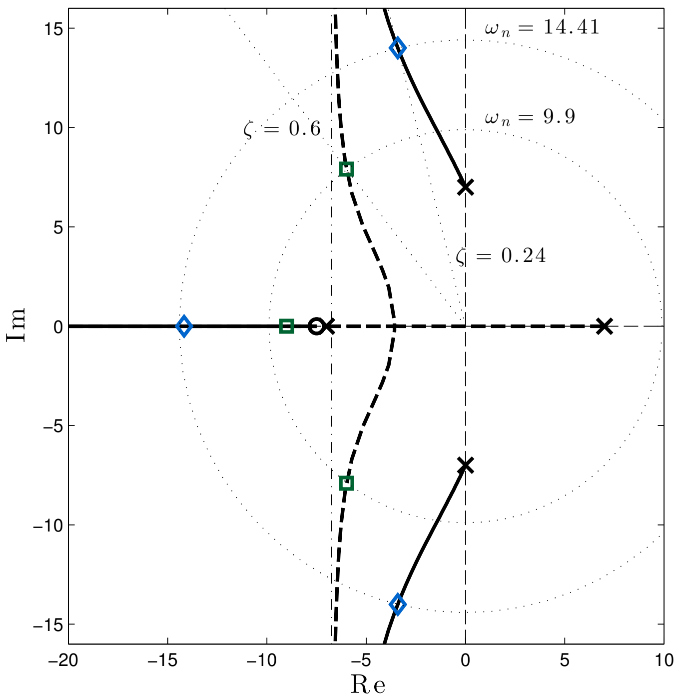


Fig. 6.19: Root-locus for the lead control (6.17) of the simple pendulum in closed-loop with the model linearized around the stable equilibrium $\bar{\theta} = 0$; root-locus for the model linearized around the unstable equilibrium $\bar{\theta} = \pi$ is shown in dashed for comparison

The lead controller was designed to achieve good performance around the unstable equilibrium $\theta = \pi$. Before proceeding we evaluate its performance also around the stable equilibrium $\theta = 0$. Fig. 6.19 shows the root-locus obtained with controller (6.17) in closed-loop with the pendulum model linearized around $\theta = 0$. The root-locus with the pendulum model linearized around $\theta = \pi$ is shown in dashed for comparison. Note that the closed-loop natural frequency is close to the one predicted before with the proportional-derivative controller, i.e. $\omega_n \approx 14.41 \approx 2\omega_n^{ol}$, but this controller displays much less closed-loop damping, $\zeta \approx 0.24$. This means that one should be careful if the same controller is to be used around both equilibria, even though, in practice, additional mechanical damping which we chose to ignore during the design might improve the overall damping. We will justify this later in § 8.1 when we perform a nonlinear analysis of the closed-loop system.

Let us turn our attention back to the issue of tracking. As discussed earlier in § 5.7, one should be careful when using a model linearized around equilibrium as the basis for a tracking controller that will necessarily attempt move

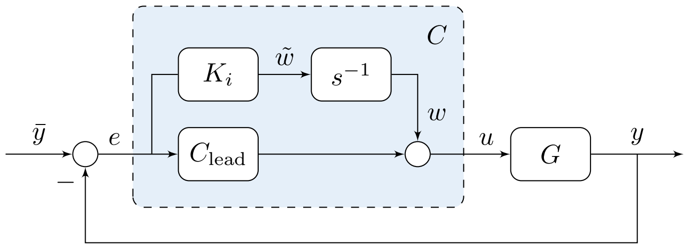


Fig. 6.20: Block-diagram for controller with additional integral action

the system away from equilibrium: if the model near the new equilibrium is too different than the one for which controller was originally designed then the design might no longer be valid. Having said that, we cautiously proceed using the model linearized around the unstable equilibrium $\theta = \pi$, ready to perform a redesign if necessary.

Our goal is to add integral action to the loop. Our previous attempt to simply add a pole at the origin, Fig. 6.15, revealed serious problems with closed-loop stability. For this reason we must take a different route. Instead of adding the integrator in *series* with the controller, we add it in *parallel*, the resulting controller being very close to an implementable PID controller, as we will see. We assume that $C_{\text{lead}}(s)$ has already been designed to provide regulation performance, our goal is to design K_i to achieve adequate levels of integral action. The feedback connection is illustrated in Fig. 6.20. The controller resulting from this diagram has the transfer-function:

$$C(s) = \frac{K_i}{s} + C_{\text{lead}}(s) = K \frac{s^2 + (z + K_i/K)s + p\, K_i/K}{s\,(s+p)} \tag{6.18}$$

which has not only an extra pole at the origin but also an extra zero, because the denominator and the numerator of the controller have both degree 2.

At first it might not be obvious what loop transfer-function $L(s)$ to choose for a root-locus analysis of the diagram in Fig. 6.20. Our goal is to visualize the closed-loop poles as a function of K_i so we compare Fig. 6.20 with Fig. 6.7 to conclude that $L(s)$ must be the transfer-function between the input $\tilde{w}$ and the error signal e. From Figs. 6.20 and 4.8, the closed-loop transfer-function from w to e has already been computed in (4.26):

$$e = D\, w, \qquad D = \frac{G^{\pi}}{1 + G^{\pi}\, C_{\text{lead}}},$$

with which

$$e = L\, \tilde{w}, \qquad L = s^{-1} D = \frac{s^{-1} G^{\pi}}{1 + G^{\pi}\, C_{\text{lead}}}.$$

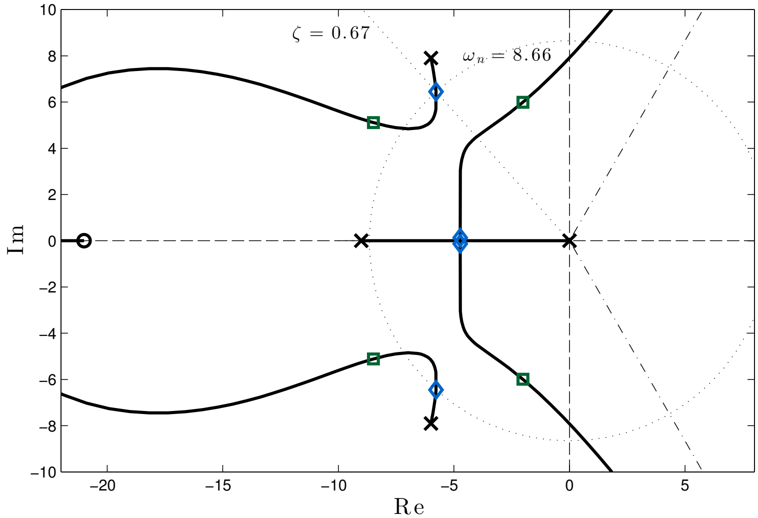


Fig. 6.21: Root-locus for the lead control with additional integral action (6.18); gain is chosen to have double real poles close to the origin and a slightly smaller natural frequency, $\omega_n \approx 8.68$, marked with diamonds; alternative choice of gain that matches $\omega_n \approx 9.90$ leads to highly underdamped poles close to the imaginary axis, marked with squares; high integral gains eventually lead to instability

It follows[8] that L has as poles the closed-loop poles of the connection of the G^π with C_{clead} plus the origin, and has as zeros the poles of C_{lead}. The corresponding root-locus is plotted in Fig. 6.21.

Intuitively, if C_{lead} is able to internally stabilize the closed-loop then a small enough K_i should still preserve stability. Too large K_i will generally lead to instability. Too small K_i however will produce a closed-loop system that has a pole too close to the origin, making the closed-loop system too slow. The presence of three asymptotes in Fig. 6.21 explains why too much integral gain will lead to instability. The real path departing from the origin explain why too little integral gain will leave a pole too close to the origin.

Roots on the path that depart from the pair of complex poles have more and more damping as K_i grows. However, it is the pair of roots departing from the real poles that will become dominant and they are the ones that need special attention. With this in mind we select as integral gain the root-locus gain:

$$\alpha = K_i \approx 1.20 \tag{6.19}$$

which corresponds to closed-loop poles at the break-away point from the real axis, marked in Fig. 6.21 by diamonds. This means the fastest possible response

[8]See if you can verify this in the root-locus diagram in Fig. 6.21!

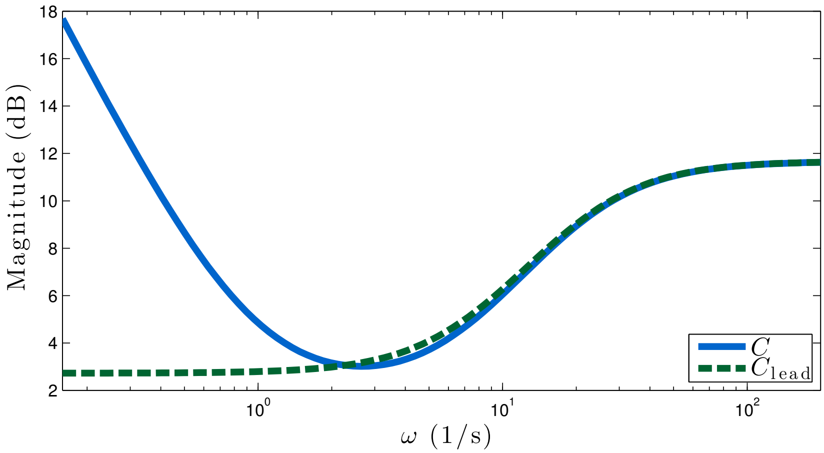


Fig. 6.22: Magnitude of the frequency response of the lead controller (6.17) and the lead controller with integral action (6.18)

from the dominant poles that is not oscillatory[9]. Note that the choice of gain $K_i \approx 2.8$ leads to the only possible stable roots that intersect the circle $\omega_n \approx 9.90$, marked in Fig. 6.21 by squares, but we do not opt for this choice because it produces a pair of highly underdamped complex poles.

The controller resulting from selecting K_i as in (6.19) has transfer-function:

$$C(s) \approx 3.83 \frac{(s + 6.86)(s + 0.96)}{s\,(s + 21)} \tag{6.20}$$

Note how the magnitude of the frequency response of the final controller C and the lead controller C_{lead} are very close in high-frequencies, Fig. 6.22. In low-frequencies, C displays the typical unbounded low-frequency gain provided by the integrator.

Of course one may take different paths to arrive at a suitable controller. For instance, one might have concluded after Fig. 6.15 that a second zero was needed and designed a second-order controller by placing an additional zero and pole and studying the resulting root-locus diagram. The diagram in Fig. 6.23 shows the root-locus one would study if a second-order controller with the same poles and zeros as $C(s)$ were selected as in (6.20). The closed-loop poles marked by diamonds recover (6.20). Note how arguments similar to the ones brought when discussing previous root-locus plots can be used to help locate the poles and zeros in this direct design. One might also enjoy the complementary discussion in § 7.8, in which we will revisit the control of the simple pendulum using frequency domain techniques.

[9] The other two poles will still contribute oscillatory components to the response.

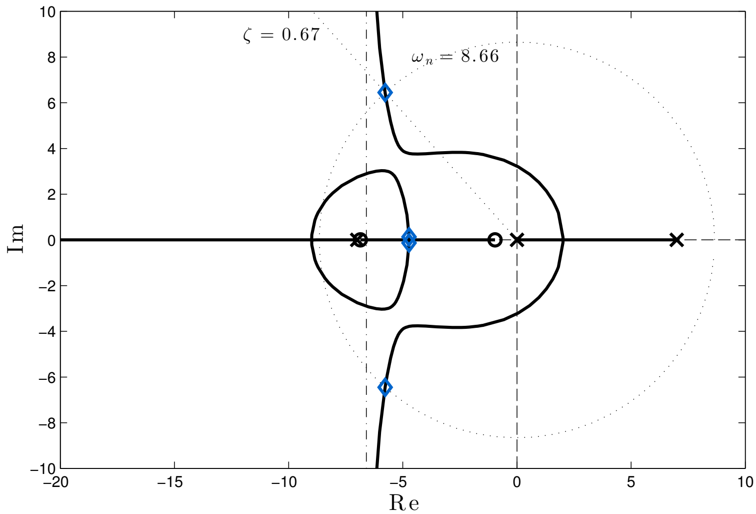


Fig. 6.23: Alternative root-locus for the direct study of the control of the pendulum in closed-loop with a second-order controller with the same poles and zeros as (6.20); diamond marks indicate choice of gain that leads to the exact same closed-loop poles as in Fig 6.21

Problems

P6.1. Show that the roots of the second-order polynomial

$$s^2 + 2\,\zeta\,\omega_n s - \omega_n^2 = 0, \qquad \omega_n > 0.$$

are always real and at least one of the roots is positive.

P6.2. Show that the step response of the second-order system with transfer-function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\,\zeta\,\omega_n s + \omega_n^2}, \qquad \omega_n > 0$$

is

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \pi/2 - \phi_d),$$
$$\omega_d = \omega_n\sqrt{1-\zeta^2},$$
$$\phi_d = \tan^{-1}\frac{\zeta}{\sqrt{1-\zeta^2}},$$

for $t \geq 0$.

P6.3. The mass-spring-damper diagram in Figure 6.26 is used to model the suspension dynamics of a car [Jaz08, Chapter 14]. The mass m represents $1/4$ of the mass of the car and m_u represents the mass of the wheel. The constants k and b are the stiffness and damping coefficient of the spring and shock absorber. The ordinary differential equation:

$$m\ddot{x} + b\dot{x} + kx = ky + b\dot{y}$$

constitute a simplified description of the motion of this model where x is a displacement measured from equilibrium. If

$$z = x - y,$$

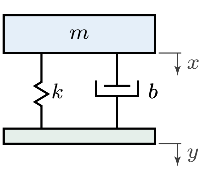
show that

$$m\ddot{z} + b\dot{z} + kz = -m\ddot{y}$$

where y is the road profile.

P6.4. Calculate the transfer-function from the road profile y to the displacement z. Calculate

Fig. 6.24: One-eight-car model, **P6.3**



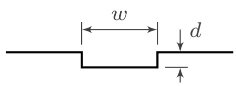


Fig. 6.25: Diagram for **P6.5**

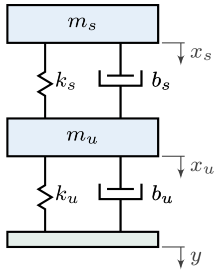


Fig. 6.26: One-quarter car model, **P6.7**

the value of the spring stiffness and shock absorber damping coefficient for a car with $1/4$ mass equal to 640kg to have a natural frequency $f_n = 10$Hz and damping ratio $\zeta = 0.08$. Locate the roots in the complex plane.

P6.5. Use MATLAB to plot the response of the car suspension with parameters as in **P6.4** to a pothole with a profile as shown in Fig. 6.25 where w =1m and d = 5cm for a car traveling at 10km/h. Repeat for a car traveling at 100km/h. Comment on your findings.

P6.6. If the road profile is $y(t) = \cos(\lambda v t)$ where v is the car's velocity, what is the worst possible velocity a car with suspension parameters as in **P6.4** can be traveling as a function of the road wavelength λ?

P6.7. The spring-mass-damper diagram in Figure 6.26 is used to model the wheel-body dynamics of a car [Jaz08, Chapter 15] and is an improved version of the model considered in **P6.3**. The mass m_s represents $1/4$ of the mass of the car without the wheels and m_u represents the mass of a single wheel. The constants k_s and b_s are the stiffness and damping coefficient of the spring and shock absorber. The constants k_u and b_u are the stiffness and damping coefficient of the tire. The ordinary differential equations:

$$m_s\ddot{x}_s + b_s(\dot{x}_s - \dot{x}_u) + k_s(x_s - x_u) = 0$$
$$m_u\ddot{x}_u + (b_u + b_s)\dot{x}_u + (k_u + k_s)x_u - b_s\dot{x}_s - k_s x_s = k_u y + b_u\dot{y}$$

constitute a simplified description of the motion of the quarter-car model where x_s and x_u are displacements measured from equilibrium. If

$$x = x_s - x_u, \qquad z = x_u - y,$$

show that

$$m_s m_u \ddot{x} + b_s(m_s + m_u)\dot{x} + k_s(m_s + m_u)x - k_u m_s z - b_u m_s \dot{z} = 0$$

$$m_u \ddot{z} + k_u z + b_u \dot{z} - k_s x - b_s \dot{x} = -m_u \ddot{y}$$

where y is the road profile.

P6.8. Calculate the transfer-function from the road profile y to the displacement $x + z$. If the tire stiffness is $k_u = 200000$N/m and the tire damping coefficient is negligible, i.e. $b_u = 0$, use MATLAB to calculate the value of the spring stiffness and shock absorber damping coefficient for a car with $1/4$ mass m_s equal to 600kg and wheel mass equal to $m_u = 40$kg to have its dominant poles have a natural frequency $f_n = 10$Hz and damping ratio $\zeta = 0.08$. Locate all the roots in the complex plane.

P6.9. Use MATLAB to plot the response of the car suspension with parameters as in **P6.8** to a pothole with a profile as shown in Fig. 6.25 where w =1m and d = 5cm for a car traveling at 10km/h. Repeat for a car traveling at 100km/h. Comment on your findings.

P6.10. If the road profile is $y(t) = \cos(vt/\lambda)$ where v is the car's velocity, what is the worst possible velocity a car with suspension parameters as in **P6.8** can be traveling as a function of the road wavelength λ?

P6.11. Compare the answers from **P6.8–P6.10** with the answers from **P6.4–P6.4**.

P6.12. We have shown in § 5.4 that the nonlinear differential equation:

$$J_r\,\ddot{y} + b\,\dot{y} + m\,g\,r\sin\theta = u$$

is an approximate model for the motion of the simple pendulum in Fig. 5.11 and that $(\bar{\theta}, \bar{u}) = (0, 0)$ and $(\bar{\theta}, \bar{u}) = (\pi, 0)$ are the pendulum equilibrium points. Calculate the nonlinear differential equation obtained in closed-loop with the linear proportional controller:

$$u = K\,e, \qquad e = \bar{\theta} - \theta.$$

Show that $(\bar{\theta}, \bar{u}) = (0, 0)$ and $(\bar{\theta}, \bar{u}) = (\pi, 0)$ are still equilibrium points. Linearize the closed-loop system linearized about $(\bar{\theta}, \bar{u}) = (0, 0)$ and $(\bar{\theta}, \bar{u}) = (\pi, 0)$ and calculate the associated transfer-function. Assuming all constants are positive, find the range of values of K that stabilize both equilibrium points.

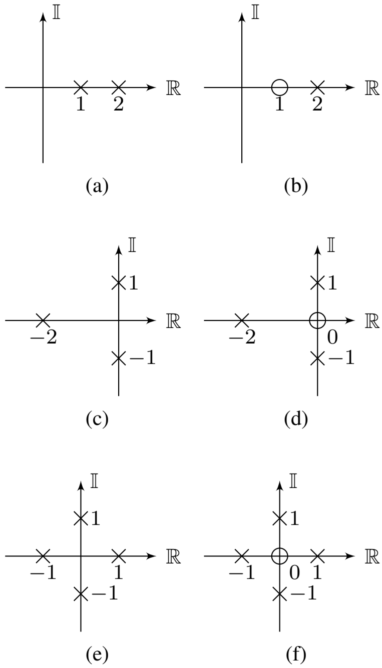


Fig. 6.27: Block diagrams for **P6.14**

P6.13. Repeat **P6.12** with the linear proportional-derivative controller:

$$u = K_p\,e - K_d\,\dot{e}, \qquad e = \bar{\theta} - \theta.$$

P6.14. First sketch the root-locus for the SISO systems with poles and zeros shown in Fig. 6.27 then use MATLAB to verify your answer.

P6.15. Find a proper feedback controller that can stabilize the SISO systems with poles and zeros shown in Fig. 6.27. Is the transfer-function of the controller asymptotically stable?

P6.16. You have shown in **P2.7** and **P2.9** that the ordinary differential equation

$$J_r\,\dot{\omega}_1 + b_r\,\omega_1 = r_2^2\,\tau, \quad \omega_2 = (r_1/r_2)\,\omega_1,$$

is a simplified description of the motion of a rotating machine driven by a belt without slip as in Fig. 2.18, where

$$J_r = J_1 r_2^2 + J_2 r_1^2, \quad b_r = b_1 r_2^2 + b_2 r_1^2,$$

ω_1 is the angular velocity of the driving shaft and ω_2 is the machine's angular velocity. Let $r_1 = 0.05$m, $r_2 = 0.25$m, $m_1 = 1$kg, $m_2 = 10$kg, $b_1 = 0.125$kg m/s, $b_2 = 6.25$kg m/s, $J_i = m_i r_i^2/2$, $i = 1, 2$. Use the root-locus method to design an I-controller:

$$\tau = K \int_0^t e(\sigma)\, d\sigma, \qquad e = \bar{\omega}_2 - \omega_2$$

and select K so that both closed-loop poles are real and as negative as possible. Is the closed-loop capable of asymptotically tracking a constant reference input $\bar{\omega}_2$?

P6.17. Redo **P6.16** using a PI-controller and select the gains such that both closed-loop poles have negative real part more negative than the pole of the machine without performing a pole-zero cancellation. Is the closed-loop capable of asymptotically tracking a constant reference input $\bar{\omega}_2$?

P6.18. The belt system in **P6.16** is connected to a piston that applies a periodic torque that can be approximated by $\tau_2(t) = h\cos(\sigma t)$, where the angular frequency σ is equal to the angular velocity ω_2. The modified equations including this additional torque is given by:

$$J_r\, \dot{\omega}_1 + b_r\, \omega_1 = r_2(r_2\, \tau_1 + r_1 \tau_2).$$

Use the root-locus method to design a controller so that the closed-loop system is capable of asymptotically tracking a constant reference input $\bar{\omega}_2(t) = \bar{\omega}_2 = 4\pi$, $t \geq 0$, and asymptotically rejecting the torque perturbation $\tau_2(t) = h\cos(\sigma t)$ when $\sigma = \bar{\omega}_2$.

P6.19. You have shown in **P2.10** that the ordinary differential equation

$$\begin{aligned} J\, \dot{\omega} + (b_1 + b_2)\, \omega &= \tau + g\, r(m_1 - m_2), \\ J &= J_1 + J_2 + r^2(m_1 + m_2), \\ v_1 &= r\, \omega, \end{aligned}$$

is a simplified description of the motion of the elevator in Fig. 2.19, where ω is the angular velocity of the driving shaft and v_1 is the elevator's load linear velocity. Let $r = 1$m, $m_1 = 1000$kg, $m_2 = 800$kg, $b_1 = b_2 = 120$kg m/s, $J_1 = J_2 = 200$kg m^2, and $g = 10$m/s^2. Use the root-locus method to design a dynamic controller $\tau = K(\bar{x}_1 - x_1)$ that can asymptotically track a constant position reference $\bar{x}_1(t) = \bar{x}_1$, $t \geq 0$, where $x_1 = \int_0^t v_1(\tau)\, d\tau$ is the elevator's load linear position.

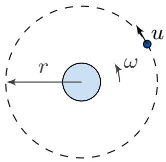


Fig. 6.28: Satellite in orbit, **P6.20**

P6.20. The differential equations:

$$\begin{aligned} m(\ddot{r} - r\,\omega^2) &= -\frac{GMm}{r^2} \\ m(2\,\dot{r}\,\omega + r\,\dot{\omega}) &= u \end{aligned}$$

are a simplified description of the motion of a satellite orbiting earth as in Fig. 6.28, where r is the satellite's radial distance from the center of the earth and ω is the satellite's angular velocity, u is the force applied by a thruster in the tangential direction, m is the mass of the satellite, $M \approx 6 \times 10^{24}$kg is the mass of the earth, and $G \approx 6.7 \times 10^{-11}$Nm^2/kg^2 is the universal gravitational constant. Represent this equation in a block-diagram using only integrators and rewrite the differential equations in state-space.

P6.21. Show that if

$$\Omega^2 R^3 = GM$$

then $u(t) = \dot{r}(t) = 0$, $r(t) = R$ and $\omega(t) = \Omega$ is an equilibrium point of the equations in **P6.20**. Linearize the equations about the equilibrium point to obtain the linearized system in state space:

$$\begin{aligned} \dot{x} &= \begin{bmatrix} 0 & 1 & 0 \\ 3\Omega^2 & 0 & 2\Omega \\ 0 & -2\Omega & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m} \end{bmatrix} u, \\ \tilde{y} &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x, \end{aligned}$$

where

$$x = \begin{pmatrix} r - R \\ \dot{r} \\ R\omega - R\Omega \end{pmatrix}, \qquad \tilde{y} = y - R.$$

Use MATLAB to calculate the transfer-function from the tangential thrust u to the output $\tilde{y}$ and design a controller using the root-locus method that can stabilize the radial distance of the satellite in closed-loop.